

Penumbra2.0

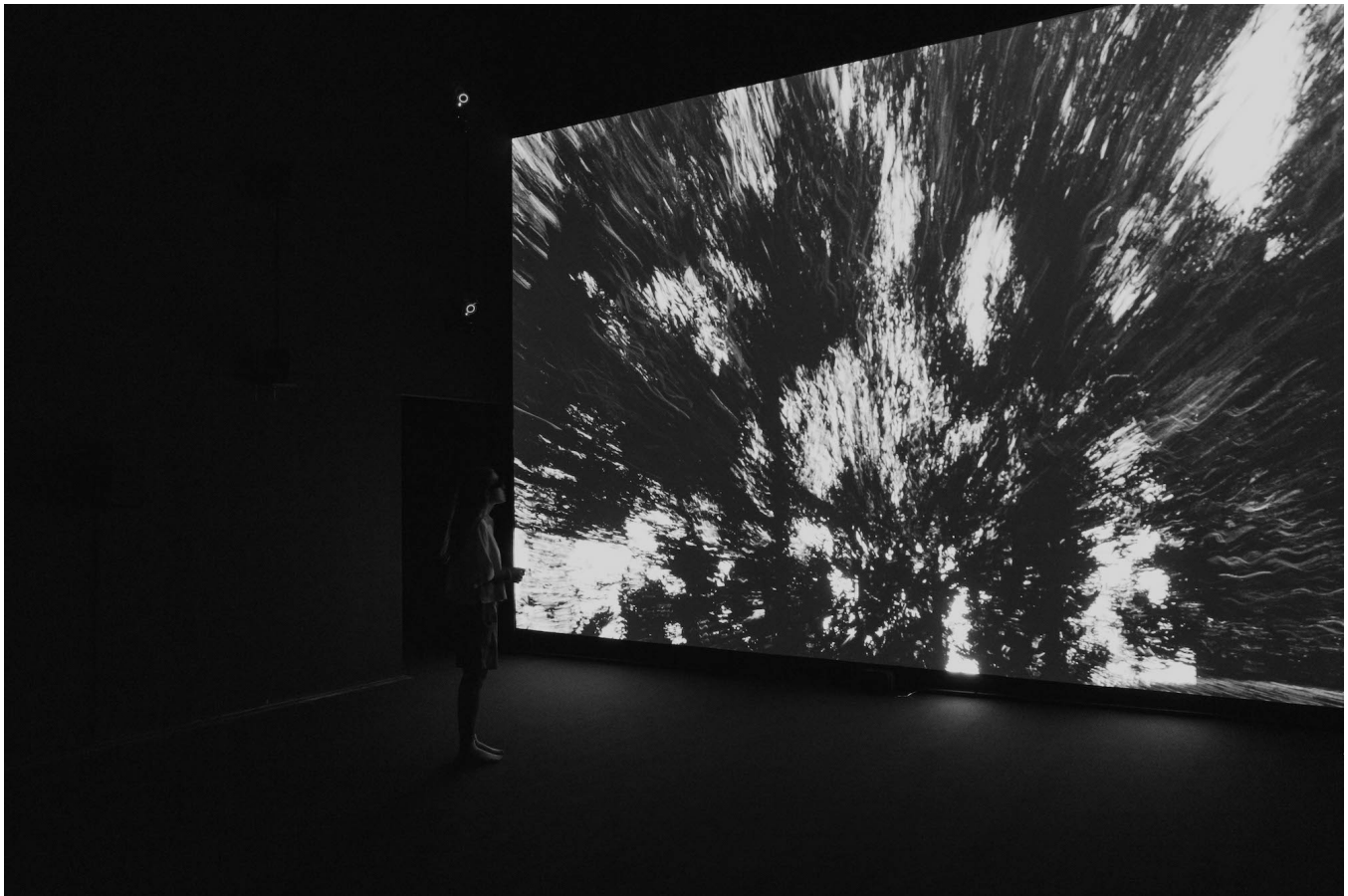
Dennis Del Favero
Yang Song
d.delfavero@unsw.edu.au
yang.song1@unsw.edu.au
UNSW iCinema Centre
Sydney, Australia

Khalid Moinuddin
khalid.moinuddin@vu.edu.au
Institute for Sustainable Industries
and Liveable Cities
Victoria University
Melbourne, Australia

Charles Green
c.green@unimelb.edu.au
University of Melbourne
Melbourne, Australia

Jason Sharples
j.sharples@adfa.edu.au
ADFA
Melbourne, Australia

Alex Ong
Navin Brohier
alex.ong@unsw.edu.au
n.brohier@unsw.edu.au
UNSW iCinema Centre
Sydney, Australia



Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Art Gallery '23, December 12-15, 2023, Sydney, NSW, Australia

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0308-9/23/12.

<https://doi.org/10.1145/3610537.3622946>

ABSTRACT

Penumbra2.0 is an AI immersive art installation exploring the visualisation of an unpredictable extreme wildfire scenario using a pyro-aesthetic, an aesthetic based on the perceptual qualities of this fire type. Currently seen in Canada, these wildfires are extreme in their speed and scale. They are unpredictable as their behaviour moves across terrains in unforeseen ways, unlike the linear and

predictable paths of bushfires. Using geo-located data, Penumbra2.0 recreates an actual wildfire in the Vosges mountains, France, 2020. It has been collaboratively developed by art, AI and fire researchers.

Penumbra2.0 forms part of a larger research program entitled iFire. iFire consists of an artistic and scientific project series, the Penumbra series comprising the artistic and Umbra the scientific. Both use the same database of atmospheres, flora, pyro-histories and topographies. Penumbra explores the palpable and sensorial qualities of wildfire experiences, while Umbra investigates the dynamic variables of wildfire events. To amplify the evocative viscosity of these encounters, Penumbra is rendered in monochrome. To underscore its complex pyro-turbulent processes, Umbra is rendered in color.

Penumbra2.0 investigates pyro-aesthetics as a two-way dialogue between the viewer and a fire-laden landscape, rather than a linear

relationship between an active human protagonist and a passive “natural disaster”. It aims to model the uncertainties that characterize such exchanges. On the one hand, the actions of the user, and on the other, the fire’s behavior. As the user traverses the landscape, they attempt to control their perspective through their movement and orientation of gaze. The fire responds in autonomous ways, by changing its behavior. Conversely, unexpected changes in the wildfire induce shifts in the user’s actions as they attempt to manage the uncertainty.

ACM Reference Format:

Dennis Del Favero, Yang Song, Khalid Moinuddin, Charles Green, Jason Sharples, Alex Ong, and Navin Brohier. 2023. Penumbra2.0. In *Proceedings of SA Art Gallery '23*. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3610537.3622946>